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| **Fiscal Year:** | FY 2025 | **Task Last Updated:** | FY 05/19/2025 |
| **PI Name:** | Moreno-Villanueva, María Ph.D. | | |
| **Project Title:** | DNA Damage in the ISS Astronaut's Lymphocytes and Their Association with Stress-Induced Immune Dysfunction (Internal Project) | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | (1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | maria.moreno-villanueva@uni-konstanz.de | **Fax:** | FY |
| **PI Organization Type:** | GOVERNMENT | **Phone:** | |
| **Organization Name:** | c/o NASA Johnson Space Center | | |
| **PI Address 1:** | 2101 NASA Parkway | | |
| **PI Address 2:** | | | |
| **PI Web Page:** | | | |
| **City:** | Houston | **State:** | TX |
| **Zip Code:** | 77058 | **Congressional District:** | 36 |
| **Comments:** | Formerly affiliated with the Radiation Biophysics Laboratory at NASA Johnson Space Center; also affiliated with the Center for Chemical Biology at the University of Konstanz (Universität Konstanz). | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2014-15 HERO NNJ14ZSA001N-MIXEDTOPICS. Appendix E: Behavioral Health & Human Health Countermeasures Topics |
| **Start Date:** | 01/01/2016 | **End Date:** | 09/30/2025 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Brocato, Becky | **Contact Phone:** | |
| **Contact Email:** | becky.brocato@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | Per NASA, Dr. Honglu Wu has retired and will no longer be Principal Investigator (PI) on the project. A Co-Investigator, Dr. María Moreno-Villanueva, will replace Dr. Wu as PI (Ed., 4/25/25). | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | Internal Project | | |
| **Performance Goal No.:** | | | |

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| **Performance Goal Text:** | |
| **Task Description:** | NOTE: This is an integrated project consisting of Dr. Brian Crucian's "Functional Immune Alterations, Latent Herpesvirus Reactivation, Physiological Stress, and Clinical Incidence Onboard the International Space Station" directed research; and Dr. Richard Simpson's "The Impact of an ISS Mission on the Anti-Viral and Functional Properties of NK-cells, T-cells, B-cells and Dendritic Cells," Dr. Kanokporn Rithidech's "Effects of Space Flights on the Proteome of Astronauts' Plasma," and Dr. Honglu Wu's "DNA Damage in the ISS Astronaut's Lymphocytes and Their Association with Stress-Induced Immune Dysfunction" solicited research.<br>The unique radiation environment encountered in space has been known to cause deleterious effects in humans, and these effects are of special concern for prolonged space missions beyond the protective terrestrial magnetosphere. DNA lesions, namely single- and double-strand breaks, are the most common of these effects and the most likely to compromise health. In addition to radiation, astronauts in space experience microgravity and other stress factors. In human cells, immune cells in particular, stress-induced catecholamine and corticoid release, as well as increased levels of inflammation, are known to induce DNA damage and impair DNA repair, which in turn results in single and double-strand breaks. Intact DNA repair machinery is crucial for immune cell functionality. Therefore understanding the interaction between DNA damage response and immune response is an important step towards improving knowledge of the mechanisms associated with immune dysfunction in the space environment.<br><br>In this study, we propose to quantify DNA damage in astronauts' lymphocytes by measuring the frequency of single- and double-strand breaks. We will also perform RNA-seq analysis with a focus on pathways involved in DNA damage response and/or inflammatory response. Measures of DNA damage obtained before, during, and after long-duration International Space Station (ISS) missions will then be compared with the mRNA levels. The outcomes in the present study will also be compared to the measured T cell function, cytokine profiles and viral reactivation in the Integrated Immune Flight Study, which is the parental project of our proposed study. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | October 2025 Update: Dr. Wu, the previous PI, will be submitting the final report for this investigation. So this record has been de-activated (Ed., 10/7/25).<br>NOTE: Continuation of "DNA Damage in the ISS Astronaut's Lymphocytes and Their Association with Stress-Induced Immune Dysfunction" with new Principal Investigator Dr. María Moreno-Villanueva, beginning April 2025. |
| **Bibliography Type:** | Description: (Last Updated: ) |